Manuel Bonilla López, PhD.

Lead Manufacturing Engineer ● MES and Automation

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Google Scholar Profile

PROFESSIONAL SUMMARY

Lead-level manufacturing and process engineer with a Ph.D. in Physics and 16+ years of experience across advanced manufacturing environments, including semiconductors and optics. Scientist-engineer by training, combining deep materials-physics insight and hands-on process expertise with enterprise-scale manufacturing systems leadership.

Architect and technical lead for multi-site digital transformation initiatives spanning manufacturing execution systems, data analytics, and operational governance. Designed and delivered a reference MES implementation now being adopted across multiple sites and divisions, supporting both semiconductor and night-vision/optics operations.

Recognized as a trusted technical leader and internal consultant who translates complex manufacturing processes into standardized, scalable digital frameworks—defining data models, workflows, governance, and performance metrics that enable real-time decision-making, operational reliability, and cost-efficient execution at the enterprise level.

AREAS OF EXPERTISE

LEADERSHIP & OPS.	Digital transformation | Multi-site MES governance | Change Review Board |Capacity & headcount planning | Budget and license optimization
MANUFACTURING	Manufacturing Execution System (MES) | Statistical Process Control (SPC) architecture | Process routes integration |Tool and product tracking | Visual Task Boards | Environmental tracking |Design of Experiments (DOE) | Root Cause and Corrective Action (RCCA)
DEPOSITION SYSTEMS	Molecular Beam Epitaxy (MBE) | Atomic Layer Deposition (ALD) | E-beam evaporation | Sputtering | Thermal Evaporation | Plasma-Enhanced Chemical Vapor Deposition (PECVD)
AUTOMATION & DATA	SharePoint | Power BI dashboards | MS teams alerts | Database management | Data models and metrics governance | GitHub
PROGRAMMING	Python | R | Microsoft VBA | SQL | HTML/JSON/CSS |MATLAB | JMP | Origin Lab
OPTICAL DESIGN	FilmStar | OpenFilters | TFCalc | McLeod
METROLOGY	Spectrophotometer | Ellipsometers | Scanning Tunneling Microscopy (STM) | X-RAY Spectroscopy (XPS) | Physical Property Measurement System (PPMS) |Atomic Force Microscope (AFM) | Scanning Electron Microscope (SEM) |Electrical characterization

PROFESSIONAL EXPERIENCE

L3Harris — Sr. MANUFACTURING ENGINEER (MES AND AUTOMATION LEAD) — MAR 2024 – Present

Operating at lead/principal scope; reporting directly to Director of Operations.

– Expanded Scope: Enterprise and Multi-Site MES Leadership | Multi -Site — NOV 2025 – Present

- Selected as reference MES architect for enterprise expansion across AZ, OH, NH, and FL.
- Leading MES architecture, governance, and deployment across semiconductor and night-vision manufacturing sites.
- Training future site MES administrators, managers, and project leads.
- Driving enterprise MES licensing alignment to reduce redundant six-figure annual site costs.

– **Foundational Scope: Department Digital Transformation |** Palm Bay**,** FL MAR 2024 – NOV 2025

- Architected and deployed the Eyelit MES as the foundational manufacturing system for the Microelectronics Department (semiconductors), serving ~200 users; acted as sole MES owner and decision authority during initial rollout.
- Established department-wide MES governance, defining naming conventions, operational hierarchies (layer vs. module), ownership rules, and required data collection standards across engineering, quality, operations, and maintenance.
- Transformed informal Word-based procedures into structured digital travelers, integrating recipes, parameters, inspections, and decision logic to standardize execution and traceability.
- Designed and institutionalized the department's first end-to-end SPC and analytics framework, authoring plans, standards, review cadence, and metrics governance; trained 100+ users and led recurring SPC reviews.
- Built integrated SharePoint and Power BI operational platforms connected to MES, enabling real-time visibility into tool status, wafer movement, WIP, and performance indicators.
- Implemented environmental health monitoring strategy, including RH/temperature logger deployment, ISO-based alarm limits, and automated Python + Task Scheduler pipelines feeding live dashboards and Teams notifications
- Served on the Change Review Board, evaluating process changes and partnering with deposition engineering to improve E-beam chamber performance, including implementation of revised crystal monitoring methods.
- Developed the facilities manpower and workload model for the department by analyzing all building-level facilities assets. Reviewed manuals, extracted required preventive maintenance tasks, and validated labor hours and frequencies through direct collaboration with shared facilities personnel. Built a year-long task calendar incorporating a two-shift operating model, no-weekend coverage, vacation and non-productive time assumptions, and a 75% utilization standard directly enabling approval and hiring of dedicated facilities personnel assigned exclusively to the department's buildings.
- Awarded the Rising Star Award by sector-level operations leadership for consistently exceeding performance expectations and delivering high-impact manufacturing technology solutions, including successful implementation of the Eyelit MES and automated visualization platforms. Recognition cited innovation, cross-functional collaboration, and long-term project adoption and sustainability across the organization.

LOCKHEED MARTIN — Sr. OPTICAL PROCESS ENGINEER | Orlando, FL JAN 2022 – JUN 2023

Led the oxide deposition manufacturing team processes supporting thin-film optical coating programs, overseeing deposition systems, yield improvement initiatives, and cross-functional coordination between operations, engineering, and vendors.

- Led the oxide thin-film manufacturing group producing antireflective and high-reflective optical filters across the visible and infrared spectrum.
- Served as manufacturing owner for PVD and CVD deposition systems from Tecport Optics and Bühler Leybold, including E-beam evaporation, sputtering, thermal evaporation, and plasma-enhanced CVD.
- Implemented structured RCCA methodologies that significantly reduced defective runs and enabled production delivery up to one month ahead of schedule for select programs for the first time in the department's history.
- Diagnosed and resolved complex process and equipment issues impacting yield and reliability, including coating delamination, non-uniformity, durability failures, spectral deviations, chamber leaks, and temperature calibration errors.
- Recovered critical manufacturing assets previously labeled out-of-production, including an environmental humidity test chamber and a PECVD system for diamond-like carbon (DLC).
- Identified systemic vendor quality gaps contributing to manufacturing failures and led corrective actions aligning supplier capabilities with program requirements.
- Awarded the Performance Management Team Advocate Award for improving communication between operators and management through implementation of a visual task board and daily issue-resolution workflows.

INTEL CORPORATION — SEMICONDUCTOR PROCESS ENGINEER | Hillsboro, OR JUN 2020 – MAY 2021

Supported high-volume semiconductor manufacturing as a front-end process engineer, owning critical deposition tools and driving equipment reliability, process stability, and technology transitions.

- Front-end manufacturing tool owner at Intel's D1 fab supporting deposition of high-k dielectric films for FinFET transistors using ALD.
- Owned two manufacturing tools comprising four deposition sub-chambers, ensuring film quality through thickness control, defect density monitoring, and statistical data trending.
- Reduced tool downtime and improved availability by optimizing preventive maintenance strategies using DOEs, pump life analysis, and chamber pressure diagnostics.
- Coordinated with equipment vendors to sustain high-volume production and resolve equipment performance issues.

- Led conversion of two tools from a previous product line to a current production flow, supporting manufacturing continuity during technology transitions.

--------------- RESEARCH & TECHNICAL EXPERIENCE ---------------

MATERIAL SCIENCE & SURFACE PHYSICS RESEARCH ASSOCIATE | USF Tampa, FL AUG 2013 – AUG 2020

Led hands-on experimental research in thin-film deposition, ultra-high vacuum systems, and advanced materials characterization, with full ownership of process development, equipment operation, and data analysis.

- Specialized in surface science and two-dimensional materials with a focus on physical vapor deposition using MBE.
- Synthesized and characterized monolayer Vanadium Selenide (VSe_2), providing the first published experimental evidence of room-temperature two-dimensional ferromagnetism, foundational to the modern 2D magnetism field. Developed and optimized deposition recipes for monolayer $TiSe_2$, $MoSe_2$, WSe_2, and $TaSe_2$ to study phase behavior, charge density waves, magnetic states, excitonic properties, band structure, doping, and surface morphology.
- Installed, configured, and maintained ultra-high vacuum systems.
- Led installation and initial commissioning of a shared departmental MBE deposition system, supporting multi-user research operations.
- Research resulted in 10 peer–reviewed publications with over 2,300 citations (Goggle Scholar Link), including a feature highlight by *Nature Nanotechnology's* own editorial article.

MATERIAL SCIENCE RESEARCH ASSOCIATE | UPR Humacao, PR AUG 2010 – AUG 2013

Conducted hands-on research in gas-sensing device fabrication, integrating polymer nanofiber deposition, microfabrication techniques, and electrical characterization to develop functional sensor devices for defense and smart-textile applications.

- Fabricated gas-sensing diodes and transistors by electrospinning polymer nanofibers directly onto silicon wafers, forming active sensing layers for chemical detection.
- Performed metal electrode deposition via thermal evaporation to define device contacts and circuit structures.
- Executed electrical characterization including I–V measurements and gas response testing to evaluate device sensitivity, stability, and performance.
- Conducted structural and morphological analysis using SEM to validate nanofiber formation and device architecture.
- Developed and optimized polymer formulations and nanofiber recipes to tune chemical composition and improve gas-sensing response characteristics.
- Supported early research toward smart-textile concepts integrating sensing devices into fabrics for environmental and defense-related gas detection.

--------------------------------- EDUCATION ---------------------------------

Ph.D., Physics, University of South Florida (USF)
M.Sc., Physics, University of South Florida (USF)
B.Sc., Physics Applied to Electrical Engineering, University of Puerto Rico (UPR)
Completed near-minor-level coursework in Computational Mathematics, University of Puerto Rico (UPR)

--------------- CERTIFICATIONS & PERSONAL PROJECT -------------

Google Data Analytics Professional Certificate | Google via Coursera
Leading People and Teams Certificate — In Progress | University of Michigan via Coursera
CS50 (Multiple Computer Science Courses) — In Progress | Harvard University via EDX
Personal Project on Finance Topics | Link via Kaggle

------------------ GUESS SCIENTIST & INTERSHIPS ------------------

GUESS SCIENTIST

- SLAC National Accelerator Laboratory **|** California, USA
- Solei Synchrotron **|** Paris, France.

INTERSHIPS

- University of Pennsylvania (UPENN)
- University of Nebraska (UNL)